COMPUTER ENGINEERING DEPARTMENT

Subject : DS **ASSIGNMENT NO – 04** **Unit – IV**

1. What is **hashing**? Explain the **properties of good hash function** with examples.?
2. What is **Hash function**. Explain any **4 types of Hashing function with example**.
3. Explain the concept of **quadratic probing using example.** What are the advantages and disadvantages of quadratic probing over linear probing?
4. What is **collision with respect to hash function** ? What are different **collision resolution techniques. Explain in details with example.**
5. Explain **Rehashing** with suitable example.
6. Difference between 1) **Open hashing and Closed hashing**
   2) **Quadratic probing and Double hashing.**
7. Write a short note on **Extendible hashing**.
8. For the given set of value **35, 36, 25, 47, 25, 01, 129, 65, 29, 16, 14, 99**. Create a hash table with **size 15** and **resolve collision using open addressing techniques.**
9. Represent the following key in memory **using Linear probing with and without replacement.** Use **modulo 10** as your hashing function.
   24, 13, 16, 15, 19, 20, 22, 14, 17, 26
10. Insert the following data in the hash table of size 10 using **linear probing with chaining without replacement :**
    131, 3, 4, 21, 61, 6, 71, 8, 9
11. For the hash table size of 10 using hash function key F(key) = key % 10 insert the following keys :
    - 65, 75, 25, 29, 85, 39, 36. **Use linear probing with chaining without replacement**
    - **25, 3, 21, 13, 1, 2, 7, 12, 4, 8 Use linear probing with replacement**
12. Assume the **size of hash table as 8**. The hash function to be used to calculate the hash value of the data **X is : X%8**. Insert the following values in hash table: **10, 12, 20, 18, 15.** Use linear probing without replacement for handling collision.

13. Insert the following data in the **hash table of size 10**, using linear probing with chaining with replacement. **Here h(x) = x%10**

21, 35, 31, 37, 32, 33, 48

14. We have a hash table of **size 10** to store integer keys, with hash function **h(x) = x mod 10**. Construct a hash table step by step using linear probing without replacement strategy and insert elements in the order **31,3,4,21,61,6,71,8,9,25**. Calculate average number of comparisons required to search given data from hash table using linear probing without replacement.

15. Insert the following data in the hash table of **size 10** using linear probing with chaining by applying with replacement : **11, 33, 20, 88, 79, 98, 68, 44, 66, 24**. Calculate average number of comparisons required to search given data from hash table.

16. For the given set of values, 76, 40, 48, 5, 55. Construct hash table with size 7 and resolve collision using **quadratic probing.**

17. Prepare hash table by inserting following elements into hash table using **Extendible Hashing:** 16, 4, 6, 22. 24, 10, 31, 7, 9, 20, 26**. Bucket Size:3**.

18. Add following keys in hash table by applying **extendible hashing mechanism**. Assume capacity of each directory to **store buckets is 3.**

Keys are 10, 20, 15, 12, 25, 30, 7, 11, 08.

********** **Best of Luck** **********